

Facile Construction of Robust Multilayered PEG Films on Polydopamine-Coated Solid Substrates for Marine Antifouling Applications

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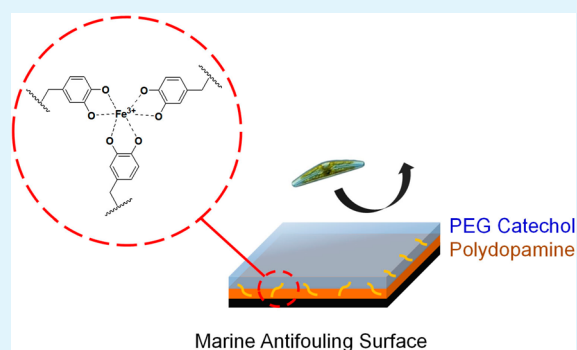
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Supporting Information

ABSTRACT: We report an effective and versatile approach to control marine fouling on artificial surfaces based on specific chemical interactions found in marine mussels. The approach consists of mussel-inspired polydopamine coating, spin-coating-assisted deposition of poly(ethylene glycol) (PEG) catechols, and their cross-linking via catechol–Fe³⁺–catechol interactions. Using this approach, multilayered PEG films that were highly resistant to marine diatom adhesion were successfully constructed on various substrates, such as stainless steel, nylon, titanium oxide, and silicon oxide. We believe that our results will provide a basis for the construction of a marine antifouling agent that can be applied by a large variety of industries owing to its applicability to different types of substrates and stability under marine environments.

KEYWORDS: polydopamine, poly(ethylene glycol), mussel inspiration, surface coating, marine antifouling



1. INTRODUCTION

Marine fouling is a universal phenomenon in which marine foulants (e.g., bacteria, microalgae, and invertebrates) attach to submerged artificial substrates.¹ Control over marine fouling has been of great interest because if left uncontrolled, it could disrupt marine ecosystems and increase fuel consumption.² Given that the surface chemical composition of solid substrates significantly affects marine fouling, efforts have focused on this aspect in hopes to suppress it.^{3,4} For instance, surface coating with functional polymers has been extensively utilized to endow marine antifouling and fouling release properties to solid substrates.^{5–31} Poly(ethylene glycol) (PEG),^{5–8} polysaccharides,^{9–11} zwitterionic polymers,^{12,13} quaternary ammonium polymers,^{24,25} and poly(2-methyl-2-oxazolines)²⁶ are examples that have been frequently used in the preparation of marine antifouling surfaces. Cao et al. grafted three types of polysaccharides (hyaluronic acid, alginic acid, and pectic acid) onto glass surfaces, via an amide bond forming reaction, and showed that alginic acid or hyaluronic acid surface immobilization effectively reduced the adhesion of *Ulva* spores.¹⁰ The zwitterionic sulfobetaine polymer-grafted glass surface, on which marine microalgae are unable to attach, also prevent marine fouling.¹² The antifouling property of zwitterion-functionalized PEG was investigated by Xu et al., who synthesized catechol- and zwitterion-bifunctionalized PEG and coated it onto solid substrates.¹⁴ An adsorption/adhesion

test of proteins, bacteria, and microalgae on the polymer-coated surface revealed its excellent antifouling property.¹⁴

Although the aforementioned polymer coatings were successfully implemented in marine antifouling applications, most studies were conducted on a specific substrate; i.e., they lacked versatility. Because various materials such as stainless steel and synthetic polymers (e.g., nylon and polyethylene) are frequently used in marine and aquatic industry, new surface coating methods that are applicable to diverse substrates are needed. Polydopamine (pDA) coatings have the potential to achieve this purpose. pDA, a mussel-inspired polymer, has been widely used in the field of surface modification because it is adherent to diverse materials; it is also possible to bind functional molecules to pDA through secondary treatments.^{32,33} Antifouling surfaces have been prepared using a pDA coating and subsequent immobilization of PEG–NH₂ and PEG–SH.³² PEG–NH₂ and PEG–SH were successfully introduced onto eight different kinds of substrates, and a dramatic reduction of protein adsorption and cell adhesion was observed.³² However, given that grafting sufficient amounts of

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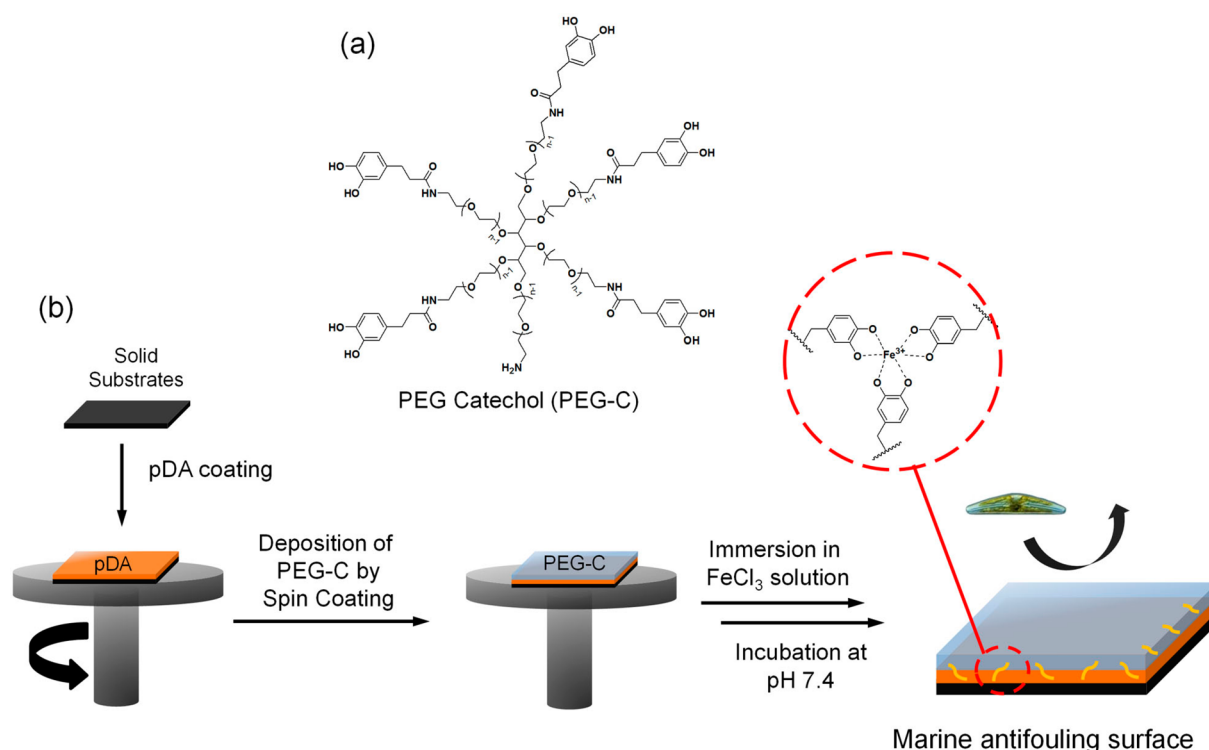


Figure 1. (a) Chemical structure of PEG-C. (b) Schematic illustration of the construction of multilayered PEG films on pDA-coated substrates.

PEG onto pDA-coated surfaces is time-consuming and pDA coating is unstable under harsh reaction conditions,³⁴ more efficient and advanced methods are required.

Recently, we reported that robust and multilayered alginate films can be easily constructed on a pDA-coated surface.³⁵ Catechol-conjugated alginates were deposited on a pDA-coated surface by spin coating and were then cross-linked with pDA via catechol- Fe^{3+} -catechol interactions. The advantage of this approach is that chemically robust and multilayered polymer films are constructed on the pDA-coated surface in a relatively short period of time (~ 4 h). We reasoned that the spin-coating-assisted deposition of polymers and subsequent cross-linking by catechol- Fe^{3+} -catechol interactions is appropriate for rapid and versatile construction of robust marine antifouling surfaces. Herein, we applied this approach for the construction of robust multilayered PEG films on diverse pDA-coated substrates. Using the approach, modification of various material surfaces (titanium oxide, silicon oxide, stainless steel, nylon, and polystyrene) was successfully carried out, and the resulting surfaces showed strong resistance against marine diatom adhesion.

2. MATERIALS AND METHODS

2.1. Materials. Dopamine hydrochloride (98%, Sigma-Aldrich), Trizma base (99%, Sigma), Trizma-HCl (99%, Sigma), iron(III) chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, 97%, Sigma-Aldrich), fibrinogen (from human plasma, Sigma), 3,4-dihydroxyhydrocinnamic acid (HCA; 98%, Aldrich), six-arm poly(ethylene glycol) amine (15 kDa MW, SunBio), 1H-benzotriazol-1-yloxytris(dimethylamino)-phosphonium hexafluorophosphate (BOP; 98%, TCI), 1-hydroxybenzotriazole monohydrate (HOBT; 97%, TCI), *N,N*-diisopropylethylamine (DIPEA; 99%, TCI), 1-methyl-2-pyrrolidone (NMP; 99%, TCI), EDTA (0.5 M, pH 8.0, Mentos), hydrochloric acid (HCl; 35–37%, Duksan), absolute ethanol (Merck), stainless steel ($\text{Fe/Cr}_{18}/\text{Ni}_{10}$; thickness, 0.05 mm; Goodfellow), nylon (thickness, 0.5 mm; size,

50 mm $\times$ 50 mm; Goodfellow), and acetone (99%, Daejung Chemicals & Metals) were used as received.

2.2. Synthesis of Six-Arm PEG Catechol. PEG catechol (PEG-C) was prepared according to a previous report.³⁶ Six-arm PEG amine (1 g) was dissolved in 10 mL of NMP at 60 °C. After complete dissolution, HCA (182.2 mg), BOP (416.3 mg), HOBT (122.52 mg), and DIPEA (139 μL) were dissolved in 5 mL of NMP and subsequently added to the six-arm PEG amine solution. The resulting mixture was vigorously stirred for 12 h at room temperature and diluted in 15 mL of 1 M HCl. The solution was transferred to a dialysis membrane (MWCO = 3500) and dialyzed for 48 h to remove unreacted coupling reagents and HCA. Acidified water obtained by adding 1 mL of 5 M HCl to 1 L of deionized water was used for the dialysis procedure. The final product was freeze-dried and stored in a refrigerator.

2.3. Polydopamine Coating. Solid substrates (1 cm $\times$ 1 cm) were cleaned with acetone or ethanol by sonication prior to use. pDA coating was carried out by immersing substrates in a buffer solution (2 mg of dopamine hydrochloride per 1 mL of 10 mM Tris, pH 8.5) at room temperature for 3 h. The coated substrates were rinsed with deionized water and dried under a stream of nitrogen gas.

2.4. Spin Coating and Fe^{3+} Coordination of PEG-C. PEG-C was deposited on pDA-coated substrates as follows. The solution of PEG-C (2 mg/mL) was dropped onto pDA-coated substrates, and the substrates were rotated at 4500 rpm for 30 s. The spin-coated substrates were then immersed in an ethanol solution of FeCl_3 (10 mM) for 5 min, followed by incubation of the resulting substrates in 10 mM PBS (pH 7.4) for 1 h to form stable catechol- Fe^{3+} -catechol bonds. Subsequently, the substrates were rinsed with deionized water and dried in a stream of nitrogen gas.

2.5. Stability Test. For the chemical stability test, PEG-C film was constructed on 9 nm thick pDA layers and incubated in an ethanol solution of FeCl_3 (100 mM) for 1 h according to the previous report.³⁴ The resulting substrates were then immersed in alkaline (10 mM NaOH) and acidic (10 mM HCl) solutions. After 1 h, the substrates were rinsed with deionized water and dried in a stream of nitrogen gas.

2.6. Diatom Adhesion. *Amphora coffeaeformis* (*A. coffeaeformis*) was obtained from the Korea Marine Microalgae Culture Center (KMMCC) and cultured in 100 mL of f/2 medium at 18 °C. A log-

phase suspension of cells was diluted with f/2 medium to give a cell suspension with a chlorophyll a content of $\sim 3.9 \mu\text{g/mL}$. A 20 mL volume of cell suspension was added to Petri dishes containing the solid substrates. After 24 h at 20 °C, the substrates were gently washed in filtered seawater to remove cells that had not attached to the surface. The attached cells were characterized and counted using an optical microscope (Nikon, LV100ND) equipped with a fluorescence-imaging system.

2.7. Characterization. X-ray photoelectron spectroscopy (XPS) was carried out using a MultiLab 2000 (Thermo VG Scientific, U.K.) with an Al K α X-ray source and ultrahigh vacuum ($\sim 10^{-10}$ mbar). The thickness of the organic layers on solid substrates was measured using an M-2000D ellipsometer (J.A. Woollam Co.). Static water contact angles were measured using a Phoenix-300 TOUCH goniometer (Surface Electro Optics Co., Ltd.). Atomic force microscopy (AFM) images were obtained in tapping mode on a Dimension Icon (Bruker, USA).

3. RESULTS AND DISCUSSION

A multilayered PEG film was constructed on solid substrates, using a three-step procedure: (1) pDA coating, (2) spin coating of PEG-C, and (3) Fe³⁺ coordination reaction (Figure 1). pDA was used as a surface primer because it can adhere to versatile substrates and introduce catechol groups onto solid substrates. The spin coating of PEG-C was carried out to deposit a multilayered PEG film on the pDA-coated surface. The resulting surface was then treated with an FeCl₃ solution to generate catechol-Fe³⁺ coordination bonds between the pDA-coated surface and PEG-C or between PEG-C molecules. The substrate with the Fe³⁺-coordinated PEG-C (PEG-C/Fe³⁺) film was further incubated in 10×10^{-3} M PBS (pH 7.4) for 1 h to form stable catechol-Fe³⁺-catechol bonds.^{37,38} Among the various solid substrates, stainless steel (SS)—a major component of marine vessels—was selected as a model substrate to test whether each step was successfully carried out.

The construction of a multilayered PEG film was characterized by water contact angle goniometry, X-ray photoelectron spectroscopy (XPS), ellipsometry, and atomic force microscopy (AFM). The water contact angle of the SS surface decreased from 70.8° to 37.2° and then to 30.8° after pDA coating and construction of PEG-C/Fe³⁺ films, indicating that the surface wettability of SS was successfully changed owing to the hydrophilic nature of pDA and PEG (Figure 2).

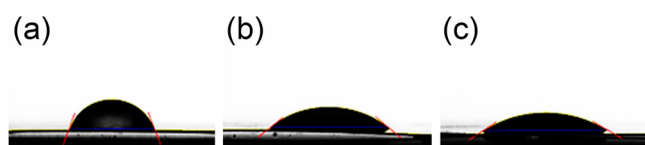


Figure 2. Water contact angle images of (a) nontreated SS, (b) pDA-coated SS, and (c) PEG-C/Fe³⁺ film.

XPS analysis provided direct evidence of surface coatings. The XPS spectrum of a pDA-coated surface showed C 1s, N 1s, and O 1s peaks, whereas no SS substrate peaks for Fe 2p and Cr 2p were detected (Figure 3a). After spin coating of PEG-C onto the pDA-coated surface, the O 1s peak increased from 19.73% to 28.78%, with a concurrent decrease in the peak intensities of C 1s and N 1s (C 1s, 73.0 $\rightarrow$ 68.1%; N 1s, 7.3 $\rightarrow$ 3.1%) (Figure 3b). After treatment of the PEG-C films with Fe³⁺, new peaks for Fe 2p were apparent (Figure 3c). For thickness and AFM analyses of PEG-C/Fe³⁺ films on pDA-coated surfaces, Si/SiO₂ substrates were used instead of SS. The coating thickness of pDA was found to be 13.1 nm. After spin coating of PEG-C,

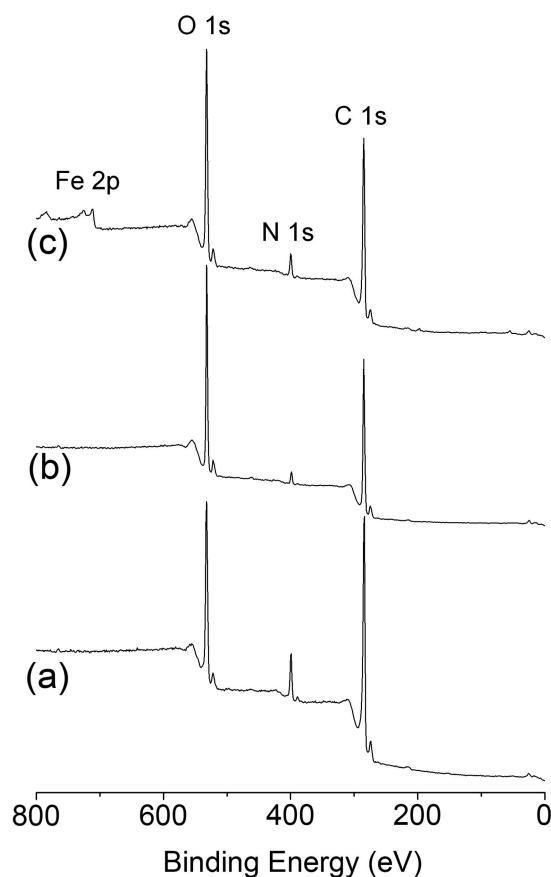


Figure 3. X-ray photoelectron spectra of (a) pDA-coated SS, (b) PEG-C film, and (c) PEG-C/Fe³⁺ film.

the coating thickness increased to 18.9 nm but decreased to 18.3 nm during Fe³⁺ treatment and incubation at pH 7.4. The results indicate 5.2 nm thick PEG-C/Fe³⁺ films on pDA-coated substrates. The AFM analysis also supported that PEG-C/Fe³⁺ film was constructed on the pDA-coated substrates. pDA coating and PEG-C/Fe³⁺ film formation made solid surfaces rougher than nontreated surfaces. The average roughness and root-mean-square roughness of pDA-coated surfaces were 9.8 and 3.2 nm and increased to 16.8 and 6.9 nm, respectively, after PEG-C/Fe³⁺ film formation (Supporting Information Figure S1).

The chemical stability of PEG-C and PEG-C/Fe³⁺ films was examined by measuring the thickness change upon chemical treatments. The PEG-C/Fe³⁺ film was stable under alkaline and acidic conditions (Figure 4a). However, the PEG-C film (no Fe³⁺ treatment) completely detached from the surface upon the same chemical treatments, and even some of the underlying pDA layer detached, too (Figure 4b). It was consistent with the chemical stability of the pDA layer, in which the layer was partly removed upon treatment of 10 mM alkali and 10 mM acid (Figure 4c).³⁴ These results revealed that Fe³⁺ plays an important role for not only grafting PEG-C on the pDA but also enhancing chemical stability of pDA and PEG-C film.

To characterize the marine antifouling property of the PEG-C/Fe³⁺ films constructed on SS surfaces, the adhesion behavior of marine diatoms was analyzed. *A. coffeaeformis*, which is frequently used in laboratory assays, was chosen as a model species.³⁹ Figure 5 shows the attached diatom density on different surfaces after being cultured for 24 h. Similar to nontreated SS, diatom adhesion occurred on the pDA-coated

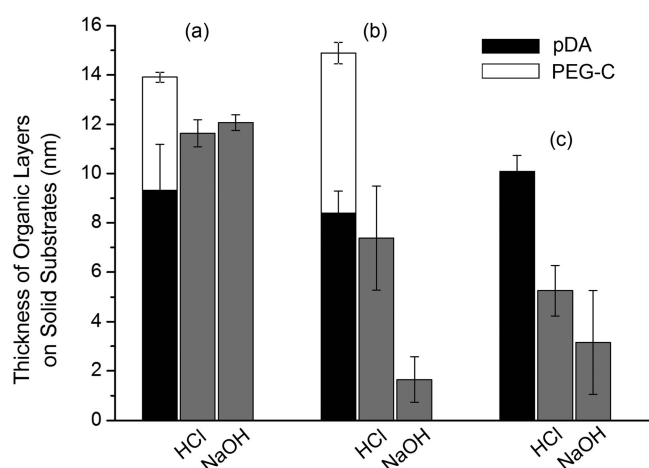


Figure 4. Thicknesses of (a) PEG-C/Fe³⁺ films, (b) PEG-C films, and (c) pDA layers before and after acid and alkali treatments.

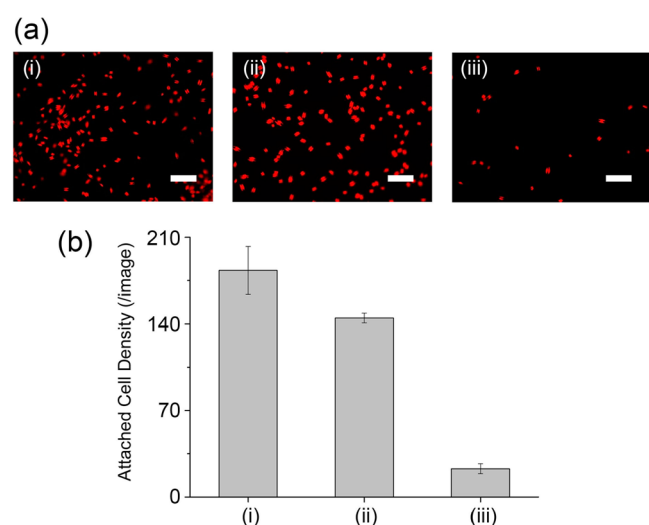


Figure 5. (a) Representative fluorescence images and (b) quantification of *A. coffeaeformis* diatoms attached on the (i) nontreated SS, (ii) pDA-coated SS, and (iii) PEG-C/Fe³⁺ film. Each point is the mean of 60 counts on three replicate samples. Error bars display the 95% confidence limits. Scale bars represent 50 μ m.

SS with a slight decrease in attached diatom density (183 $\rightarrow$ 145 diatoms/image). The change in attached diatom density is thought to result from the enhancement of surface wettability by pDA coating.⁸ In contrast, only $\sim$ 20 diatoms were attached to the PEG-C/Fe³⁺ films. Quantitatively, PEG-C/Fe³⁺ films reduced diatom adhesion by 87%, compared with the nontreated SS. These results indicate that antifouling PEG films were successfully prepared on the surface without a significant deterioration of the intrinsic properties of PEG during the process.

The advantages of our approach to constructing PEG films on surfaces are its applicability to many diverse materials by virtue of pDA coating.³² To investigate the versatility of our approach, PEG-C/Fe³⁺ films were further formed on various substrates including nylon, polystyrene (PS), silicon/silicon dioxide (Si/SiO₂), and titanium/titanium dioxide (Ti/TiO₂). Figure S2 shows that the static water contact angles of PEG-C/Fe³⁺ films on various substrates were roughly similar ($\sim$ 30°), even for substrates with fairly different initial contact angles,

suggesting that the films were successfully constructed on all the tested substrates.

The marine antifouling performance of PEG-C/Fe³⁺ films on versatile substrates was examined in the same manner as that on SS. Unlike the nontreated substrates, on which *A. coffeaeformis* heavily adhered for 24 h, PEG-C/Fe³⁺ films showed excellent antifouling performance regardless of the underlying substrate (Figure S3). The average reduction of attached cell density was more than 80% in all cases (Figure S4), and the value was comparable with that of PEG-C/Fe³⁺ films on SS. These results clearly show that virtually all substrates can be modified by this approach.

For wider applications of PEG-C/Fe³⁺ films, long-term stability under marine environments should be guaranteed, because marine vessels and equipment are typically immersed in seawater for a long time. To evaluate the long-term marine antifouling property, PEG-C/Fe³⁺ films were pretreated with seawater. After immersion for 1, 2, 3, and 4 weeks in seawater, the substrates were taken out and the diatom adhesion assay was carried out. As shown in Figure 6, PEG-C/Fe³⁺ films showed similar marine antifouling performance in the first 2 weeks. After 3 or four week immersion, an increase in attached diatom density was observed, but the PEG-C/Fe³⁺ films still

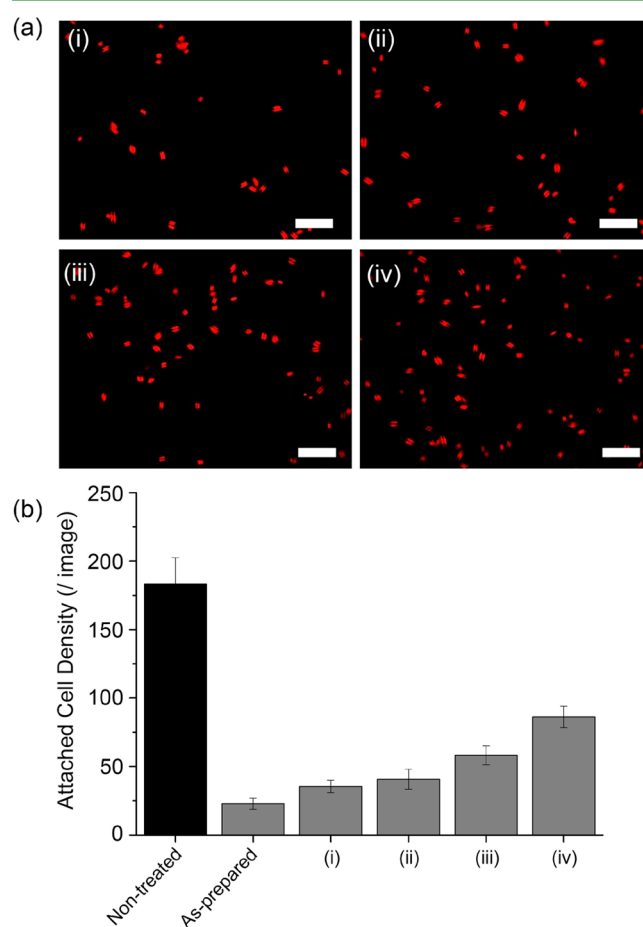


Figure 6. Stability of PEG-C/Fe³⁺ films in seawater. (a) Representative fluorescence images and (b) quantification of *A. coffeaeformis* diatoms attached on the PEG-C/Fe³⁺ film after immersion in seawater for (i) 1, (ii) 2, (iii) 3, and (iv) 4 weeks (substrate, SS). Each point is the mean of 60 counts on three replicate samples. Error bars display the 95% confidence limits. Scale bars represent 50 μ m.

showed 50% reduction of diatom adhesion, compared with nontreated SS. Given the stability of catechol- Fe^{3+} -catechol interactions even under harsh alkaline conditions,³⁵ the decrease in antifouling performance is not due to the cleavage of the interaction, but rather, it could originate from the oxidative degradation of PEG in the presence of oxygen and transition metal ions.⁴⁰ The reduced performance of PEG-C/ Fe^{3+} films in seawater could also be related to the cleavage of hydrogen bonds. In our previous work, we showed facile PEGylation on a tannic acid-coated surface by hydrogen bond formation between polyphenol and PEG.⁸ Such interaction might also play a role in constructing PEG-C/ Fe^{3+} films on pDA-coated surfaces, given the structural similarity of pDA with tannic acid. Consequently, the dissociation of hydrogen bonds between PEG and the pDA-coated surface in seawater would lead to deterioration in its antifouling performance.

4. CONCLUSIONS

In summary, multilayered PEG films were constructed on various surfaces through an integrative mussel-inspired approach for marine antifouling applications. Specifically, catechols were introduced to both solid surfaces and PEG, using a mussel-inspired pDA coating and catechol conjugation reaction, respectively. Subsequently, the catechols were cross-linked by Fe^{3+} -coordination reactions, thereby fixing PEG to the surface. Using this approach, ~5 nm thick PEG films that show excellent resistance to diatom adhesion were successfully constructed on diverse substrates. Moreover, the antifouling performance of the film was retained even after immersion in seawater for 4 weeks. Given the versatility of this approach and robustness of the prepared films, this method can serve as a universal platform for the modification/functionalization of materials to be used in marine environments.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.7b07199.

AFM images of nontreated Si/SiO₂, pDA-coated Si/SiO₂, and PEG-C/ Fe^{3+} film and representative optical images and quantification of *A. coffeaeformis* diatoms attached on various surfaces before and after PEG-C/ Fe^{3+} film formation (PDF)

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Notes

The authors declare no competing financial interest.

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